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A Note on

A toy model of mortgage market frictions

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1. Economic question and literature

Housing is both a durable source of services and an asset that can secure a loan. A rise in its price increases owners' wealth, but also changes the amount that a constrained household can borrow. The key question is whether a housing price movement reflects stronger housing demand, easier collateral conditions or changing income prospects. These mechanisms need not have the same effects on consumption and borrowing, which makes a joint interpretation of housing, credit and expenditure essential.

Kiyotaki and Moore [1997] explain how collateral values and borrowing capacity interact. Iacoviello [2005] incorporates housing and heterogeneous discount factors into a monetary business-cycle model, using impulse responses to discipline its parameters. Iacoviello and Neri [2010] develop an estimated model with a housing production sector and distinct trends. The present exercise retains patient savers, impatient borrowers and housing collateral, while holding the aggregate housing stock fixed. Housing transactions then reallocate a durable asset between groups rather than create a construction boom. Positive depreciation still requires replacement expenditure.

Gerali et al. [2010] add banking frictions and loan-rate adjustment. Iacoviello [2015] develops financial amplification through intermediaries and borrowers. These contributions help separate collateral services from a bank lending spread. In the baseline below, competitive intermediaries imply a common real borrowing and saving rate.

A binding constraint is a local regime assumption. Guerrieri and Iacoviello [2015] provide a piecewise-linear solution method with endogenously changing constraint regimes, not an exogenous Markov-switching process. Guerrieri and Iacoviello [2017] study the resulting asymmetries and estimation using an inversion filter. Their work motivates checking the scope of a local linear approximation. The present assessment keeps that approximation and asks students to discipline its empirical application rather than supplying a nonlinear estimation solution.

2. The model

All prices are in consumption-good units and R_t is a gross real return chosen at t for repayment at $t+1$. The housing stock H_t^s is held at the end of t by group $s \in \{P, I\}$. Households trade at price q_t, and housing depreciates at rate δ. Labor L_t^s is distinct from housing H_t^s.

2.1 Patient households

Patient households maximize

$$\max_{\{C_{t+\tau}^P, d_{t+\tau}, H_{t+\tau}^P, L_{t+\tau}^P\}_{\tau \geq 0}} E_t \sum_{\tau \geq 0} \beta_P^\tau \left[\varepsilon_{t+\tau}^C \log C_{t+\tau}^P + j \log H_{t+\tau}^P - \varepsilon_{t+\tau}^H \chi_P \frac{(L_{t+\tau}^P)^{1+\sigma_H}}{1+\sigma_H} \right]. \tag{1}$$

The parameter $j > 0$ values housing services. The labor-supply disturbance ε_t^H changes the marginal disutility of work, as in estimated models such as Smets and Wouters [2003]. It is not a housing preference shock. The consumption preference disturbance is ε_t^C. The budget is

$$C_t^P + q_t[H_t^P - (1-\delta)H_{t-1}^P] + d_t = R_{t-1} d_{t-1} + w_t^P L_t^P - T_t^P. \tag{2}$$

Let $\mathcal{B}_t^P$ denote this budget's income minus expenditure and U_t^P the utility in square brackets. Then

$$\mathcal{L}_t^P = E_t \sum_{\tau \geq 0} \beta_P^\tau [U_{t+\tau}^P + \lambda_{t+\tau}^P \mathcal{B}_{t+\tau}^P].$$

The first-order conditions are

$$\lambda_t^P = \varepsilon_t^C / C_t^P, \tag{3}$$

$$\lambda_t^P = \beta_P R_t E_t \lambda_{t+1}^P, \tag{4}$$

$$q_t \lambda_t^P = j / H_t^P + (1-\delta)\beta_P E_t(q_{t+1}\lambda_{t+1}^P), \tag{5}$$

$$\varepsilon_t^H \chi_P (L_t^P)^{\sigma_H} = \lambda_t^P w_t^P. \tag{6}$$

Housing yields a current service and a depreciated resale value. Deposits yield a financial return. The two Euler conditions value these different benefits.

2.2 Impatient households

Impatient households have $\beta_I < \beta_P$ and borrow against their own housing:

$$R_t b_t \leq m \varepsilon_t^M E_t(q_{t+1}) H_t^I. \tag{7}$$

The parameter m is the baseline loan-to-value ratio. The positive disturbance ε_t^M changes borrowing capacity. The baseline imposes equality locally. The relevant collateral is H_t^I, not the patient household's asset. Impatient households maximize

$$\max_{\{C_{t+\tau}^I, b_{t+\tau}, H_{t+\tau}^I, L_{t+\tau}^I\}_{\tau \geq 0}} E_t \sum_{\tau \geq 0} \beta_I^\tau \left[\varepsilon_{t+\tau}^C \log C_{t+\tau}^I + j \log H_{t+\tau}^I - \varepsilon_{t+\tau}^H \chi_I \frac{(L_{t+\tau}^I)^{1+\sigma_H}}{1+\sigma_H} \right] \tag{8}$$

subject to collateral and the budget

$$C_t^I + q_t[H_t^I - (1-\delta)H_{t-1}^I] + R_{t-1} b_{t-1} = b_t + w_t^I L_t^I. \tag{9}$$

Let $\mathcal{B}_t^I$ be income minus expenditure and $\mathcal{C}_t^I = m\varepsilon_t^M E_t(q_{t+1})H_t^I/R_t - b_t$. The Lagrangian is

$$\mathcal{L}_t^I = E_t \sum_{\tau \geq 0} \beta_I^\tau [U_{t+\tau}^I + \lambda_{t+\tau}^I \mathcal{B}_{t+\tau}^I + \lambda_{t+\tau}^I \varphi_{t+\tau}^I \mathcal{C}_{t+\tau}^I].$$

The multiplier $\varphi_t^I \geq 0$ measures the consumption value of relaxing the constraint. Its sign follows from writing collateral capacity minus debt. The conditions are

$$\lambda_t^I = \varepsilon_t^C / C_t^I, \tag{10}$$

$$1 - \varphi_t^I = \beta_I R_t E_t(\lambda_{t+1}^I / \lambda_t^I), \tag{11}$$

$$q_t = \frac{j}{H_t^I \lambda_t^I} + (1-\delta)\beta_I E_t\left[q_{t+1}\frac{\lambda_{t+1}^I}{\lambda_t^I}\right] + \varphi_t^I \frac{m\varepsilon_t^M}{R_t} E_t(q_{t+1}), \tag{12}$$

$$\varepsilon_t^H \chi_I (L_t^I)^{\sigma_H} = \lambda_t^I w_t^I. \tag{13}$$

The third term in the housing condition is its collateral service. Removing it would treat a constrained borrower as if housing did not relax borrowing.

2.3 Firms

Competitive firms combine both labor types:

$$Y_t = \varepsilon_t^A (L_t^P)^\lambda (L_t^I)^{1-\lambda}. \tag{14}$$

Here $\lambda \in (0,1)$ is the patient-labor exponent, not a population weight in the consumption identity. Firms solve

$$\min_{L_t^P, L_t^I} \; w_t^P L_t^P + w_t^I L_t^I,$$
$$\text{subject to } Y_t = \varepsilon_t^A (L_t^P)^\lambda (L_t^I)^{1-\lambda}.$$

Their Lagrangian is

$$\mathcal{L}_t^F = w_t^P L_t^P + w_t^I L_t^I + \varrho_t [Y_t - \varepsilon_t^A (L_t^P)^\lambda (L_t^I)^{1-\lambda}]. \tag{15}$$

Perfect competition in the consumption-good market gives $\varrho_t = 1$, hence

$$w_t^P = \lambda Y_t / L_t^P, \tag{16}$$

$$w_t^I = (1-\lambda) Y_t / L_t^I. \tag{17}$$

2.4 Financial intermediaries

Deposits finance loans, $d_t = b_t$. The payoff on a portfolio chosen at t is

$$\Pi_{t+1}^B = R_t^b b_t - R_t d_t. \tag{18}$$

Competition implies $R_t^b = R_t$. The constraint, not a bank spread, generates the wedge in the impatient household's Euler condition.

2.5 Equilibrium conditions

The original fixed-stock closure is

$$C_t = C_t^P + C_t^I, \tag{19}$$
$$Y_t = C_t + I_t^H, \tag{20}$$
$$I_t^H = H_t^I - (1-\delta)H_{t-1}^I + H_t^P - (1-\delta)H_{t-1}^P, \tag{21}$$
$$H_t^I + H_t^P = \bar{H}. \tag{22}$$

Thus $I_t^H = \delta\bar{H}$, not zero. Only its deviation from steady state is zero. The baseline treats aggregate replacement as a fixed real resource cost. To make this closure consistent with household purchases valued at q_t, one accounting implementation is a replacement agency whose net surplus $(q_t - 1)\delta\bar{H}$ is rebated to patient households. In the patient budget, $T_t^P = -(q_t - 1)\delta\bar{H}$. This explicit lump-sum accounting convention reconciles the resource constraint and private budgets without changing the impatient budget or the supplied fixed-stock closure.

2.6 The friction in steady state

The patient Euler condition gives $\beta_P\bar{R} = 1$. The impatient condition then gives

$$\bar{\varphi}^I = 1 - \beta_I\bar{R} = 1 - \beta_I/\beta_P > 0.$$

A positive multiplier establishes the local binding regime. It does not prove that collateral always binds globally. The price normalization is $\bar{q} = 1$. Housing stocks and consumption must be positive under the calibration.

3. Simulation and empirical starting point

The file `mortgage_RBC.mod` is a log-linear model. Lower-case code variables denote log deviations, including the positive multiplier. In particular, linearizing the borrowing Euler equation gives

$$-\frac{\bar{\varphi}^I}{1-\bar{\varphi}^I}\widehat{\varphi}_t^I = E_t\widehat{\lambda}_{t+1}^I - \widehat{\lambda}_t^I + \widehat{R}_t.$$

The collateral-service coefficient in the housing equation is $\bar{\varphi}^I m/\bar{R}$. The impatient budget also contains $\delta\bar{H}^I\widehat{q}_t$, because steady-state replacement expenditure is positive. These terms follow from the level conditions above.

The calibration is $\beta_I = .95$, $\beta_P = .99$, $\lambda = .75$, $j = .0924$, $m = .85$, $\sigma_H = 1$ and $\delta = .02$. Both labor supplies are 1/3 in steady state. Productivity persistence is .95, and the other three persistence parameters are .8. For $s \in \{A, M, H, C\}$, the level shocks follow $\log\varepsilon_t^s = \rho_s \log\varepsilon_{t-1}^s + \eta_t^s$, whose first-order counterparts are used in the code. The four unit innovation standard deviations are response normalizations, not estimated volatilities.

The downloader provides raw US output and manufacturing weekly hours. Manufacturing hours are not a housing measure. Select genuine house-price or mortgage-credit data before estimating a housing mechanism. Document real versus nominal prices, debt stocks, interest-rate annualization and common quarterly dates. The starter does not contain estimation-ready observations, a Bayesian specification or posterior answers.

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